

MAKERERE UNIVERSITY

Team AVRON

MID SEMESTER PRESENTATION

Biomedical Engineering Design 1

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The Team



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Presentation Roadmap

01

Needs finding and Screening Process

From 100 raw needs down to 3 prioritised Capstone projects.

02

Three Selected Needs

Thermal tracking, obstacle detection for the blind, & cervical dilation.

03

Progress & Next Steps

Validation plan and benchtop prototyping timeline.

NEEDS FINDING PROCESS

How we identified strategic areas of focus, carried out background research, explored clinical settings, identified needs, and drafted need statements

1. Choosing the strategic focus areas.

Need exploration:

2. Background research
3. Observation
4. Interviews
5. Need statement development

Strategic focus areas

01

Maternal & neonatal child healthcare

Target Clinician

- NICU nurse
- Pediatrician
- Gynecologist

02

Laboratory

Target Clinician

- Laboratory technologist

03

Surgery

Target Clinician

- General surgeon

04

Rehabilitation

Target Clinician

- Biomedical Engineer
- Orthopedic Surgeon

Background research & interview guides



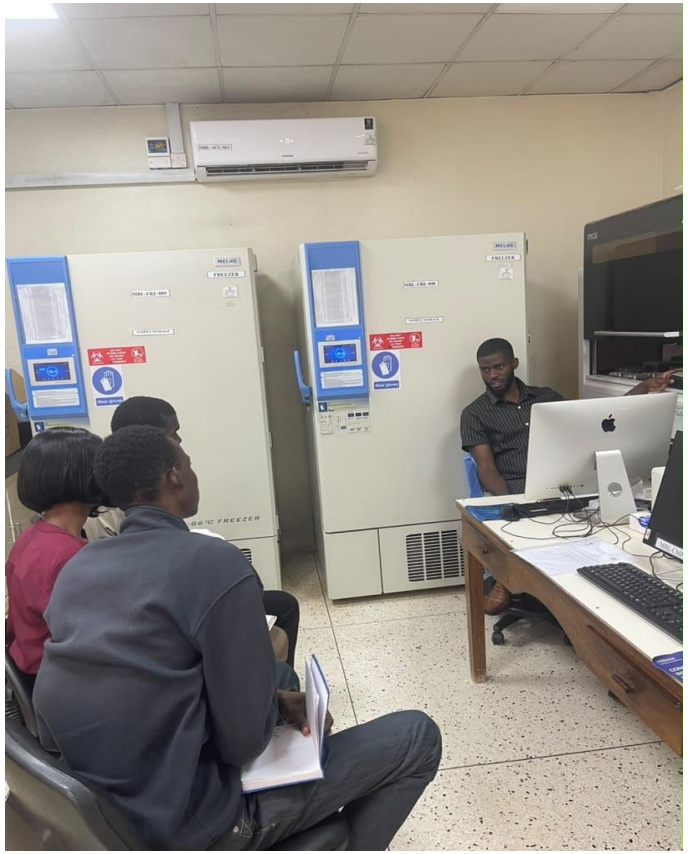
WHAT WE DID

- Reviewed background information relevant to the selected clinical areas.
- Built an initial understanding of the care environment and existing workflows.
- Drafted interview guides to keep conversations focused while allowing follow-up questions.
- Captured early questions and possible needs to investigate in the field.



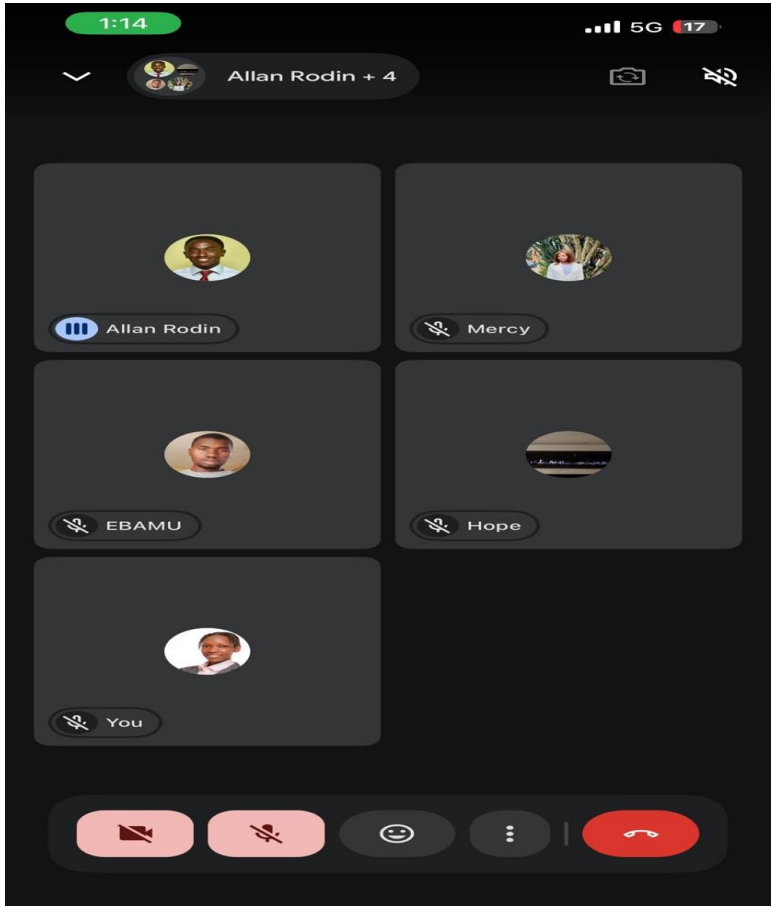
Interviews & observations





CLINICIAN INTERVIEWS
Laboratory Technologist : *Laymond Kalema*

Midwife : *Nakalanzi Sauds at Kawempe National Referral Hospital*



Biomedical Engineer : *Eng. Allan Rodin Gidoi at Ruby Hospital*

Drafting and categorizing needs

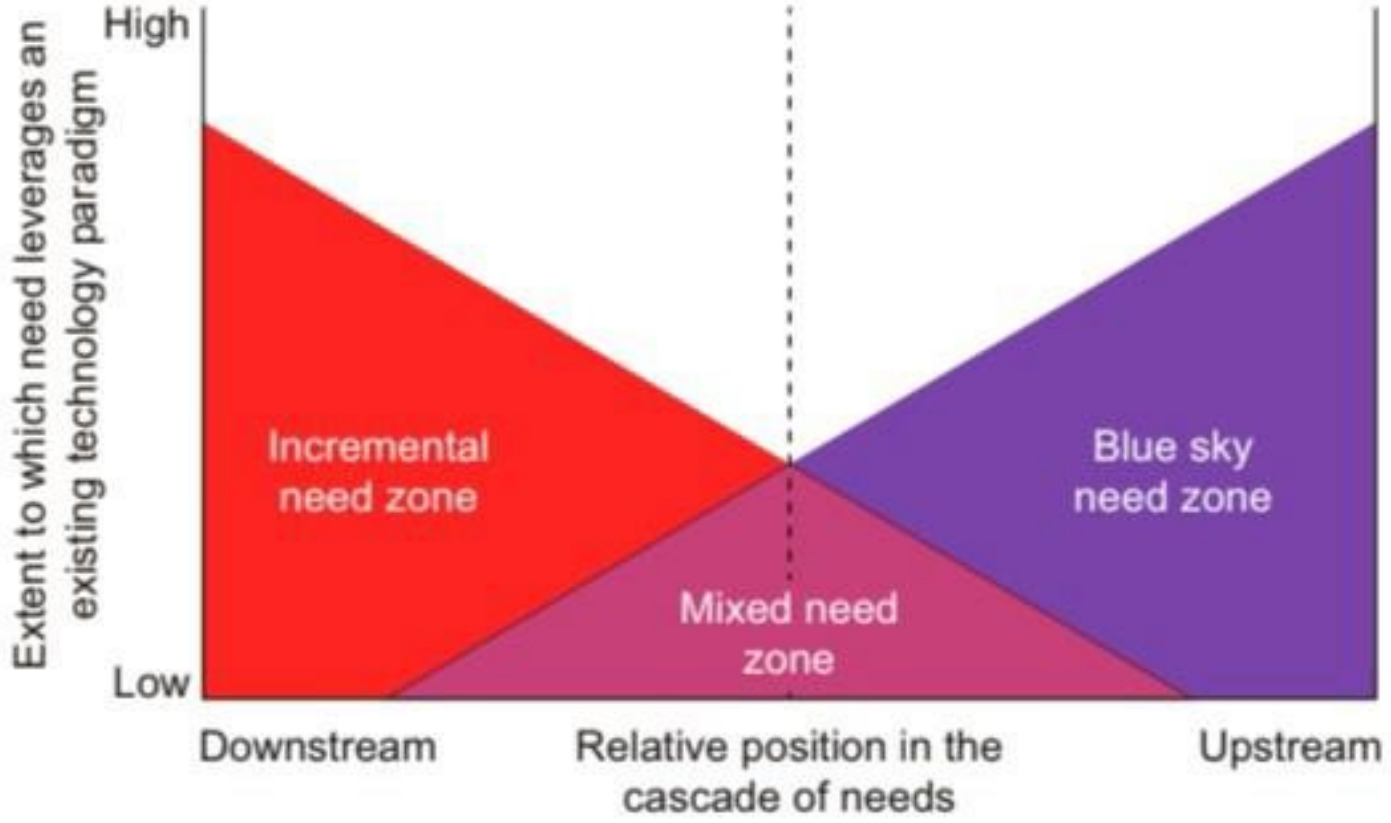
Organized the findings before writing the final need statements

NEED CLUSTERS

Group related observations and recurring challenges.

MAIN NEEDS

Highlight the strongest needs noticed across visits.



Drafting need statements

100

need statements

From our
needs-finding process

Components of a need statement

POPULATION

The person or group experiencing the problem directly

PROBLEM

What they need to do, achieve or manage

PRIMARY OUTCOME

The situation or outcome that makes the need important

Screening at a Glance

100	Needs Identified in Fieldwork	
70	Filter 1: Regulatory and Safety Rules	−30 Knocked Out
50	Filter 2: Prototyping & Technical Feasibility	−20 Knocked Out
20	Filter 3: Stanford 2x2 Opportunity Matrix	−30 Non-Disruptors
10	Filter 4: MAUT Weighted Screening Matrix	−10 Scored Out

3 FINAL SELECTION
Top Needs Selected

Regulatory, Safety & Clinical Viability

KNOCKOUT RULE

Discard Class III medical devices, multi-year clinical trial requirements, hazardous radiation, genetic manipulation, or nation-wide health policy changes.

SCREENING CRITERIA

- High regulatory classification (FDA Class III)
- High voltage, ionising radiation, or deep invasive tissue contact
- High ethical hurdles & extensive clinical trials

Prototyping & Technical Feasibility

KNOCKOUT RULE & WEIGHTS

Discard needs requiring microfluidic assays, imported chemical reagents, complex optics, or long clinical validation cycles.

Prototyping Feasibility

Benchtop Validation

Local Supply Chain

Pre-clinical Complexity

Stanford 2x2 Opportunity Matrix

Disruptors

High Importance · Low Satisfaction

Must-Haves

Market already saturated

Cost Savers

Minor incremental clinical gains

Delighters

Nice-to-have, low importance

MAUT Weighted Screening Matrix

Score = 0.20 Mkt + 0.25 Urg + 0.15 Prv + 0.25 Feas + 0.05 Saf + 0.10 LMIC

ID	Retained Need	Mkt	Urg	Prv	Feas	Saf	LMIC	Score/5.0
17	**Neonatal Temp. Monitoring**	**4.8**	**4.7**	**4.2**	**4.5**	**4.8**	**4.8**	**4.59**
28	**Obstacle Detection for the Blind**	**4.5**	**4.7**	**4.5**	**4.2**	**4.5**	**4.2**	**4.54**
90	Respiratory Distress Monitor	4.8	4.7	4.5	4.0	4.5	4.5	**4.44**
1	**Cervical Dilation**	**4.8**	**4.8**	**4.8**	**4.0**	**4.2**	**4.5**	**4.48**
38	Neonatal Sepsis Screening	4.8	4.9	4.5	3.8	4.5	4.5	**4.41**
39	Stroke Survivors Rehab Tool	4.2	3.8	4.0	4.5	4.8	4.0	**4.18**
44	Neonatal Hypoglycaemia	4.5	4.8	4.2	3.5	4.2	4.2	**4.17**
23	Surgical Trainers	3.8	3.0	4.2	4.8	5.0	4.0	**4.00**
14	Postpartum Haemorrhage	4.8	4.8	4.5	3.0	4.5	4.2	**4.15**
37	POC Cardiac Screening	4.2	4.5	4.0	3.8	4.5	4.0	**4.11**

Our Top Three Prioritised Needs

01 Neonates requiring thermal care need continuous temperature monitoring to prevent severe hypothermia & thermal stress during hospital stay.

02 People who are blind or visually impaired need a way to detect obstacles beyond the reach of a white cane to travel safely and independently.

03 Women in active labour need a non-invasive method to measure cervical dilation accurately to enable timely delivery and emergency decisions.

Neonatal Thermal Monitoring

57% – 85%

Hypothermia prevalence in sub-Saharan newborn units on admission (WHO target: 36.5–37.5°C).

4.66x Risk

Increased mortality risk associated with unmonitored neonatal hypothermia in LMICs.

CURRENT CLINICAL GAP

Intermittent spot checks fail to catch drop in body temp during radiant warmer failure or transfer between wards.

ENGINEERING OPPORTUNITY

Wearable, continuous thermal monitoring skin-sensor with visual alert when infant drops below 36.5°C normothermia.

Sources: *WHO Essential Newborn Care Guidelines (2021); Frontiers in Pediatrics (2024); Mullany et al., Journal of Pediatrics.*

Obstacle Detection Beyond White Cane Reach

43.3M

People worldwide were blind in 2020, mostly in low- and middle-income countries.

13%

Of blind and legally blind adults hit head-level obstacles monthly or more; 23% of such accidents have medical consequences.

~1 m

Reach of a white cane, which misses head- and knee-level obstacles.

CURRENT CLINICAL GAP

The white cane only senses obstacles at ground level within about 1 m, so overhanging hazards such as tree branches, poles, and signs go undetected.

ENGINEERING OPPORTUNITY

Low-cost ultrasonic add-on that warns by vibration of head- and knee-level obstacles beyond 1 m (smart canes cost ~US$200).

Sources: *Manduchi & Kurniawan (2011), Insight: Research & Practice in Visual Impairment & Blindness 4(2); Kim & Cho (2013), Int. J. of Design 7(1); GBD 2019 Blindness & Vision Impairment Collaborators, Lancet Glob Health (2021); WHO World Report on Vision (2019).*

Intrapartum Cervical Dilation Measurement

8% Deaths

Maternal deaths in Uganda directly linked to obstructed labor & delayed interventions.

Subjective PV

Digital vaginal exams have ±2 cm examiner variation and increase risk of intrapartum infection.

CURRENT CLINICAL GAP

Partograph tracking relies on repeated manual exams; lack of objective progression monitoring leads to delayed C-section decisions.

ENGINEERING OPPORTUNITY

Affordable, non-invasive acoustic or sensor-assisted measurement tool to track labor progress accurately and safely.

Sources: *WHO Labor Care Guide (2024); PLOS ONE Perinatal Care Study Uganda (2022); UNCST Audit Studies on Partograph Completion.*

Progress and Next Steps

COMPLETED SCREENING MILESTONES

- **Needs Finding:** 100 needs gathered from local clinical observations.
- **Multi-Tier Filtering:** Applied Regulatory, Technical, & 2x2 Matrix filters.
- **Weighted Scoring:** Evaluated top 20 candidates on 1-year Capstone fit.
- **Top 3 Selected:** Thermal Tracking, Obstacle Detection, & Cervical Dilation.

NEXT STEPS

- **Clinical Validation:** Interview obstetricians & neonatologists.
- **Concept Generation:** Brainstorming & morphological matrix.
- **Final Design Choice:** Select single project for design synthesis.
- **Benchtop Testing:** Build preliminary proof-of-concept prototype.


Thank you
for listening